

## Test Passes by module parts:

This screenshot shows the test results for the `AlertFactoryTest` module. The test suite consists of 5 tests, all of which passed successfully. The total execution time for the tests was 35 ms. The console output shows the command `"C:\Program Files\Java\jdk-24\bin\java.exe" ...` and the message `Process finished with exit code 0`.

| Test Name                    | Duration |
|------------------------------|----------|
| testBloodOxygenAlertFactor   | 29 ms    |
| testECGAlertFactory()        | 2 ms     |
| testBloodPressureAlertFactor | 1 ms     |
| testHypotensiveHypoxemiaA    | 2 ms     |
| testTriggeredAlertFactory()  | 1 ms     |

This screenshot shows the test results for the `AlertGeneratorTest` module. The test suite consists of 13 tests, all of which passed successfully. The total execution time for the tests was 88 ms. The console output shows the command `"C:\Program Files\Java\jdk-24\bin\java.exe" ...` and the message `Process finished with exit code 0`. The output also includes several log messages related to a TCP server and patient data.

| Test Name                          | Duration |
|------------------------------------|----------|
| testSaturationDrop()               | 33 ms    |
| testSeveralSystolicAlerts()        | 3 ms     |
| testHypotensiveHypoxemia()         | 2 ms     |
| testCriticalLowDiastolic()         | 2 ms     |
| testSaturationCriticalLowAndDrop() | 44 ms    |
| evaluationSecondTest()             | 4 ms     |
| testIncreaseDiastolic()            | 4 ms     |
| testSaturationCriticalLow()        | 4 ms     |
| testSaturationNoDrop()             | 4 ms     |
| testECGPeak()                      | 4 ms     |
| testCriticalHighDiastolic()        | 4 ms     |
| evaluationFirstTest()              | 4 ms     |
| testDecreaseDiastolic()            | 4 ms     |

This screenshot shows the test results for the `AlertStrategyTest` module. The test suite consists of 15 tests, all of which passed successfully. The total execution time for the tests was 53 ms. The console output shows the command `"C:\Program Files\Java\jdk-24\bin\java.exe" ...` and the message `Process finished with exit code 0`.

| Test Name                           | Duration |
|-------------------------------------|----------|
| testBloodPressureAlertStrategyForDe | 43 ms    |
| testBloodPressureAlertStrategyForDe | 43 ms    |
| testECGAlertStrategyNoAlert()       | 2 ms     |
| testECGAlertStrategyPeak()          | 1 ms     |
| testBloodPressureAlertStrategyForIn | 1 ms     |
| testBloodPressureAlertStrategyForDe | 1 ms     |
| testBloodPressureAlertStrategyForDe | 1 ms     |
| testBloodOxygenAlertStrategy        | 4 ms     |
| testBloodOxygenAlertStrategyHigh()  | 4 ms     |
| testBloodPressureAlertStrategyForDi | 4 ms     |
| testBloodOxygenAlertStrategyNoAle   | 4 ms     |
| testBloodPressureAlertStrategyForSy | 4 ms     |
| testBloodPressureAlertStrategyForIn | 4 ms     |
| testTriggeredAlertStrategy()        | 2 ms     |
| testBloodPressureAlertStrategyForSy | 2 ms     |

TestDecorators (alerts) 29 ms

- ✓ testPriorityAlertDecorator() 25 ms
- ✓ testRepeatedAlertDecorator() 4 ms

✓ 2 tests passed 2 tests total, 29 ms

"C:\Program Files\Java\jdk-24\bin\java.exe" ...

Process finished with exit code 0

CardioGeneratorTest (cardio\_g 37 ms

- ✓ ECGDataGeneratorTest() 31 ms
- ✓ SystolicDataGeneratorTest() 2 ms
- ✓ BloodSaturationDataGeneratc 2 ms
- ✓ DiastolicDataGeneratorTest() 1 ms
- ✓ BloodPressureDataGeneratorTest()
- ✓ BloodLevelsGeneratorTest() 1 ms

✓ 6 tests passed 6 tests total, 37 ms

"C:\Program Files\Java\jdk-24\bin\java.exe" ...

Process finished with exit code 0

DataStorageTest (data\_manage 50 ms

- ✓ testAddAndGetRecords() 39 ms
- ✓ testRemoveOldRecords() 1 ms
- ✓ testGetNonExistingPatientRec 1 ms
- ✓ testGetAllPatients() 1 ms
- ✓ testRetrieveNonExistingPatier 3 ms
- ✓ testdeletedStaffMemberDataF 1 ms
- ✓ testDataRetrieveForStaff() 1 ms
- ✓ testGetAllRecordsForPatient() 1 ms
- ✓ testLowLevelStaffDataReque: 2 ms
- ✓ testAddPatientRecord()
- ✓ testAuditLoggerAddRequest()

✓ 11 tests passed 11 tests total, 50 ms

"C:\Program Files\Java\jdk-24\bin\java.exe" ...

Process finished with exit code 0

ParserTest (dataAccess) 33 ms

- ✓ JSONParserTestWithoutTim 29 ms
- ✓ excelParserTestWithoutHead 2 ms
- ✓ brokenCSVdata() 1 ms
- ✓ JSONParserTestWithTimeStampL()
- ✓ brokenandCorrectCSVdata()
- ✓ JSONParserTestBrokenData()
- ✓ excelParserTestWithHeader() 1 ms

✓ 7 tests passed 7 tests total, 33 ms

"C:\Program Files\Java\jdk-24\bin\java.exe" ...

Process finished with exit code 0

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C:\Program Files\Java\jdk-24\bin\java.exe ...
TCP Server started on port 1
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
java.net.BindException: Create breakpoint : Address already in use: bind
    at java.base/sun.nio.ch.Net.bind0(Native Method)
    at java.base/sun.nio.ch.Net.bind(Net.java:511)
    at java.base/sun.nio.ch.Net.bind(Net.java:500)
    at java.base/sun.nio.ch.NioSocketImpl.bind(NioSocketImpl.java:630)
    at java.base/java.net.ServerSocket.bind(ServerSocket.java:326)
    at java.base/java.net.ServerSocket.<init>(ServerSocket.java:222)
    at java.base/java.net.ServerSocket.<init>(ServerSocket.java:138)
    at com.c cardio_generator.outputs.TcpOutputStrategy.<init>(TcpOutputStrategy.java:27)
> <70 folded frames>
java.net.BindException: Create breakpoint : Address already in use: bind
    at java.base/sun.nio.ch.Net.bind0(Native Method)
    at java.base/sun.nio.ch.Net.bind(Net.java:511)
    at java.base/sun.nio.ch.Net.bind(Net.java:500)
    at java.base/sun.nio.ch.NioSocketImpl.bind(NioSocketImpl.java:630)
    at java.base/java.net.ServerSocket.bind(ServerSocket.java:326)
    at java.base/java.net.ServerSocket.<init>(ServerSocket.java:222)
    at java.base/java.net.ServerSocket.<init>(ServerSocket.java:138)
    at com.c cardio_generator.outputs.TcpOutputStrategy.<init>(TcpOutputStrategy.java:27)
> <70 folded frames>

Process finished with exit code 0

```

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C:\Program Files\Java\jdk-24\bin\java.exe ...

Process finished with exit code 0

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